In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
data=pd.read_csv('https://raw.githubusercontent.com/dphi-official/Datasets/maste
```

In [3]:
```
data
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450
...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376

891 rows × 12 columns

In [4]:
```
data.shape
```

Out[4]:
```
(891, 12)
```

In [5]:
```
data.describe()
```

Out[5]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.00
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.00
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.00
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

In [6]:
```
data.describe(include='object')
```

Out[6]:

	Name	Sex	Ticket	Cabin	Embarked
count	891	891	891	204	889
unique	891	2	681	147	3
top	Braund, Mr. Owen Harris	male	347082	B96 B98	S
freq	1	577	7	4	644

In [7]:
```
data.isnull().sum()
```

Out[7]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [8]:
```
data['Age'] = data['Age'].fillna(np.mean(data['Age']))
```

In [10]:
```
data['Cabin'] = data['Cabin'].fillna(data['Cabin'].mode()[0])
```

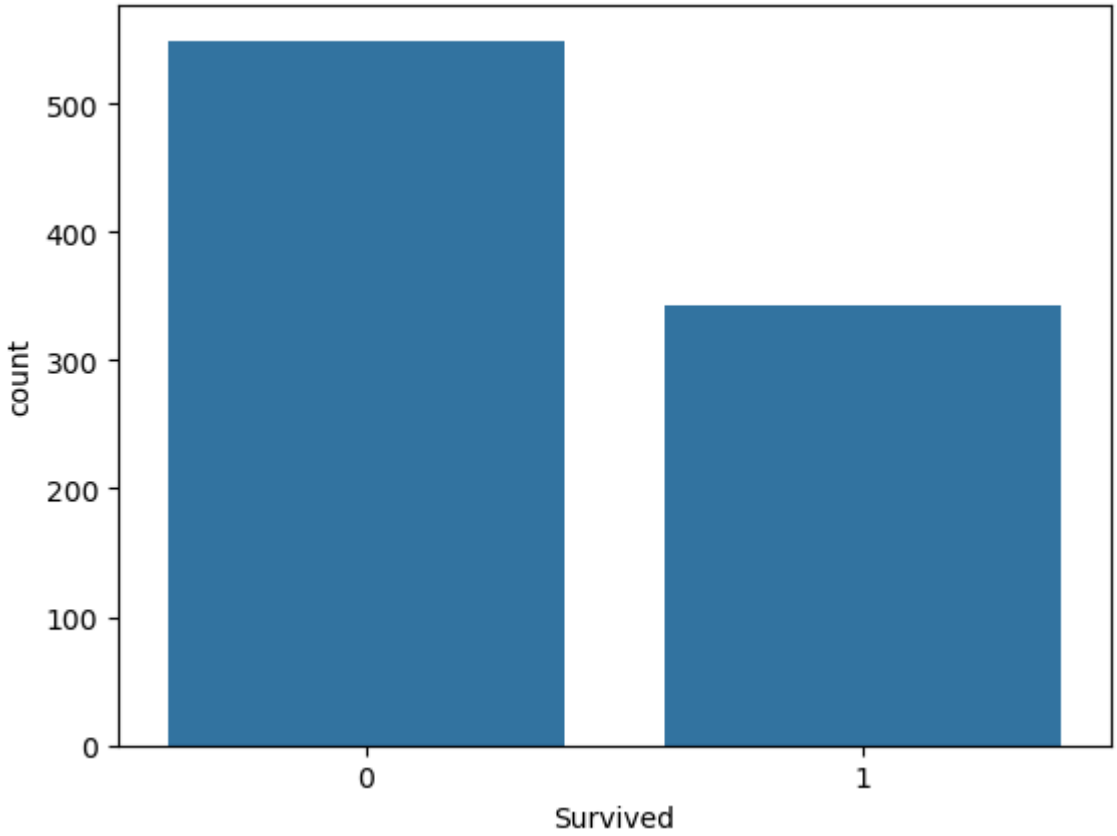
In [11]:
```
data['Embarked'] = data['Embarked'].fillna(data['Embarked'].mode()[0])
```

In [12]:
```
data.isnull().sum()
```

Out[12]:
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

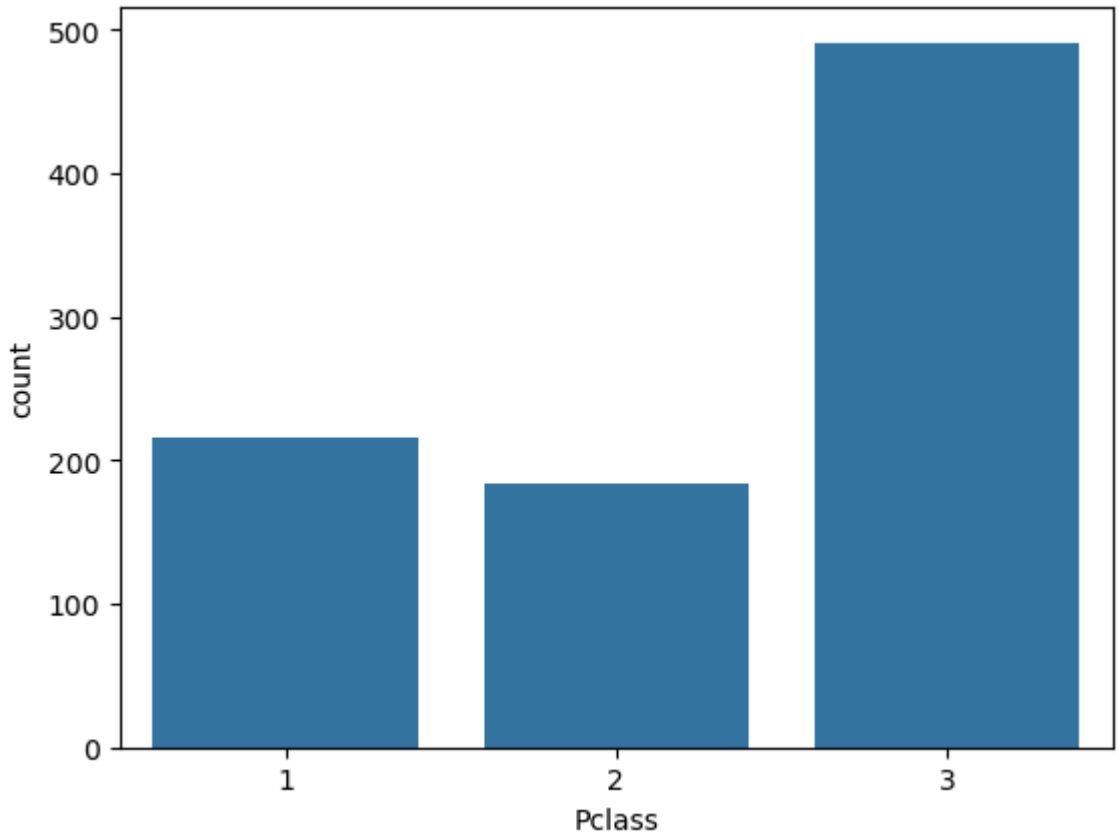
In [18]:
```
sns.countplot(x='Survived', data=data)
```

Out[18]:
```
<Axes: xlabel='Survived', ylabel='count'>
```



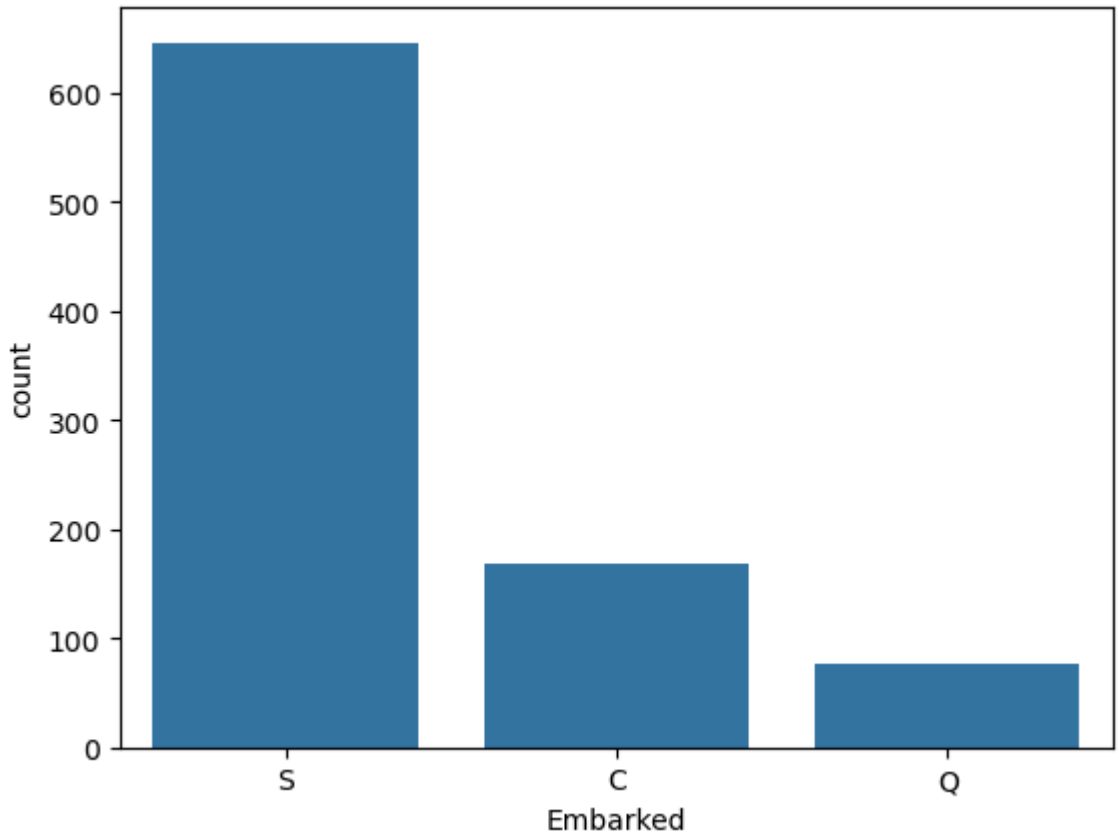
In [20]:
```
sns.countplot(x='Pclass',data=data)
```

Out[20]:
```
<Axes: xlabel='Pclass', ylabel='count'>
```



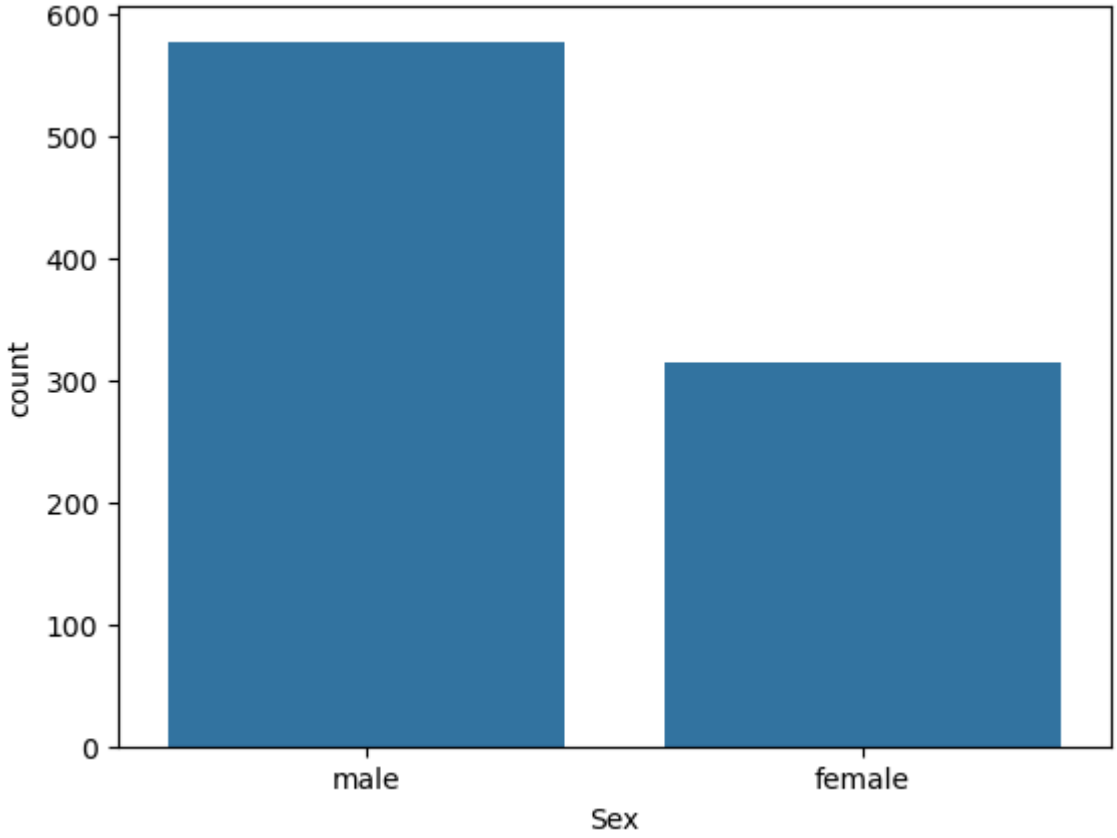
In [23]:
```
sns.countplot(x='Embarked',data=data)
```

Out[23]: <Axes: xlabel='Embarked', ylabel='count'>



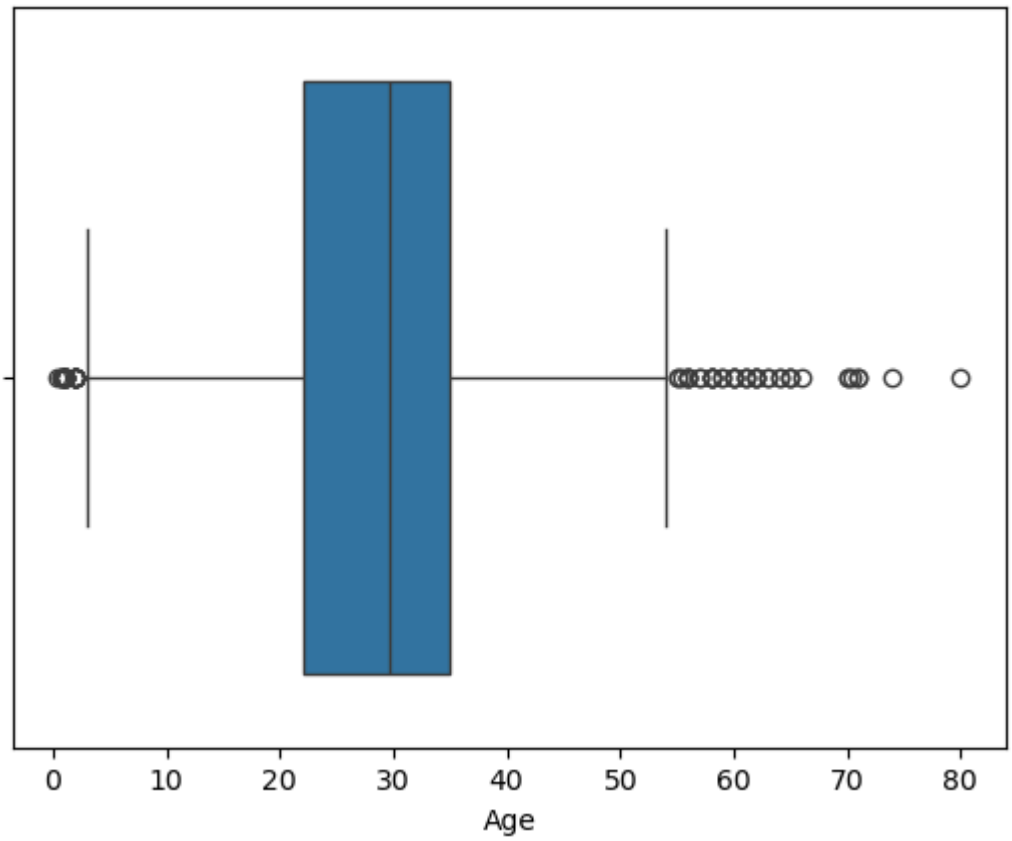
In [24]:
```
sns.countplot(x='Sex', data=data)
```

Out[24]: <Axes: xlabel='Sex', ylabel='count'>



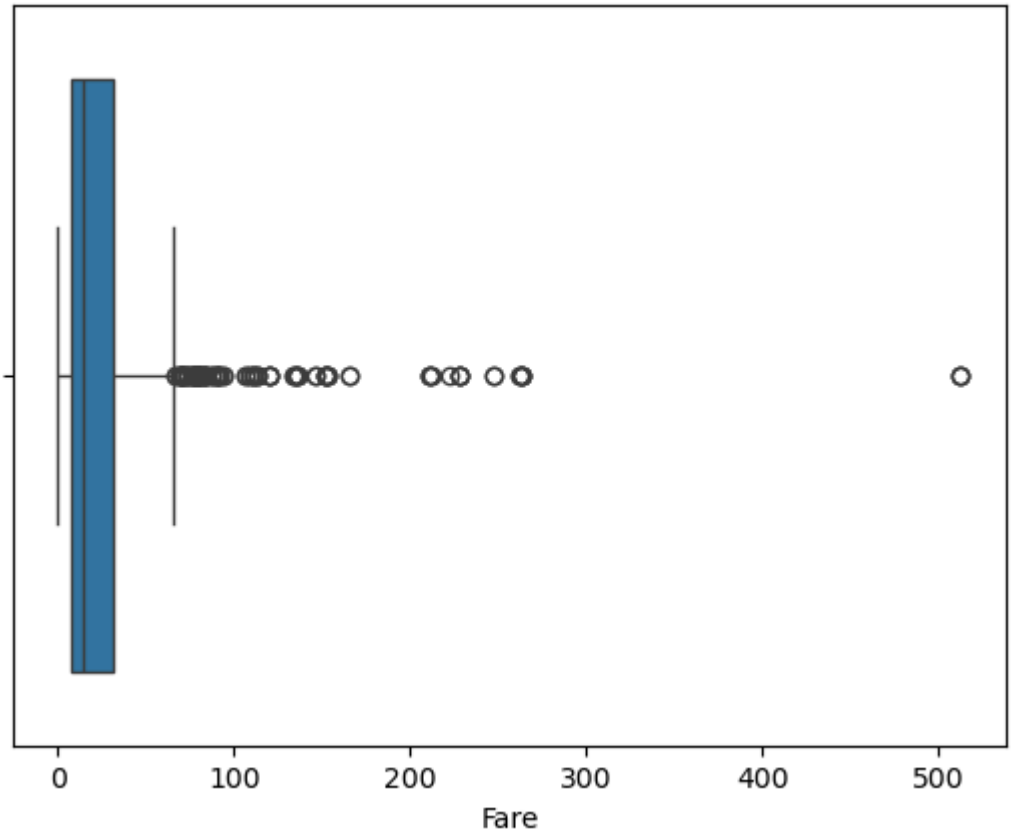
In [25]:
```
sns.boxplot(x='Age', data=data)
```

Out[25]: <Axes: xlabel='Age'>



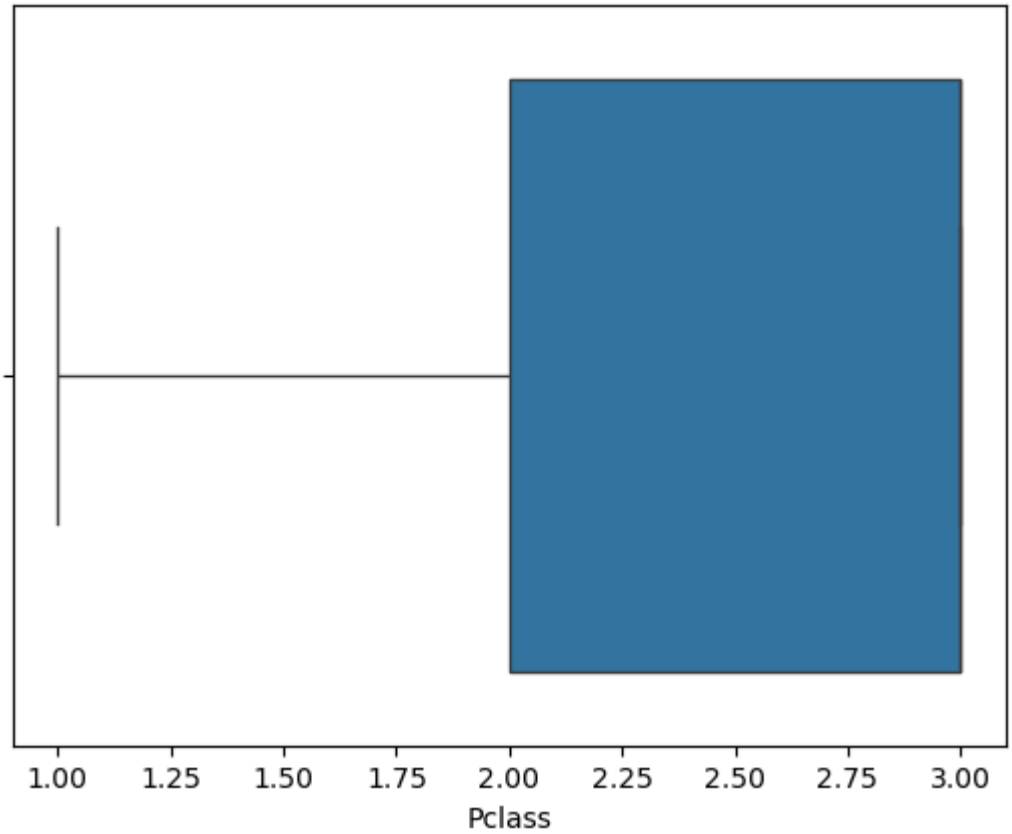
In [28]:
```
sns.boxplot(x='Fare', data=data)
```

Out[28]: <Axes: xlabel='Fare'>



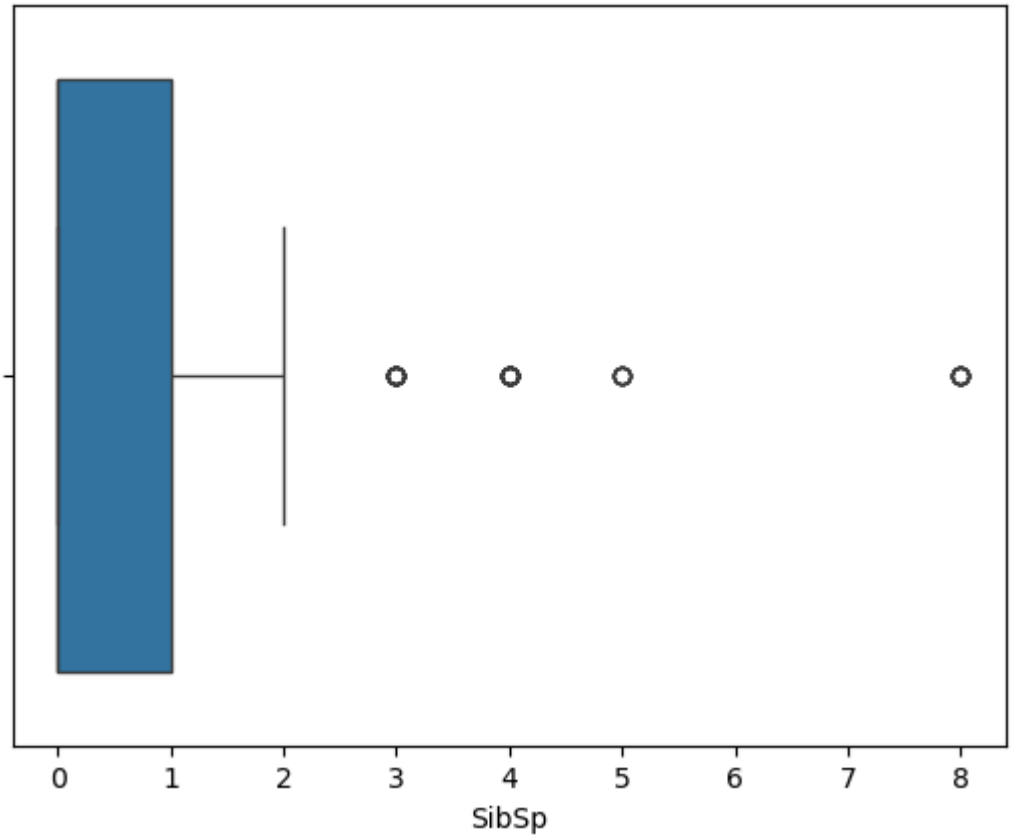
In [29]:
```
sns.boxplot(x='Pclass', data=data)
```

Out[29]:
```
<Axes: xlabel='Pclass'>
```



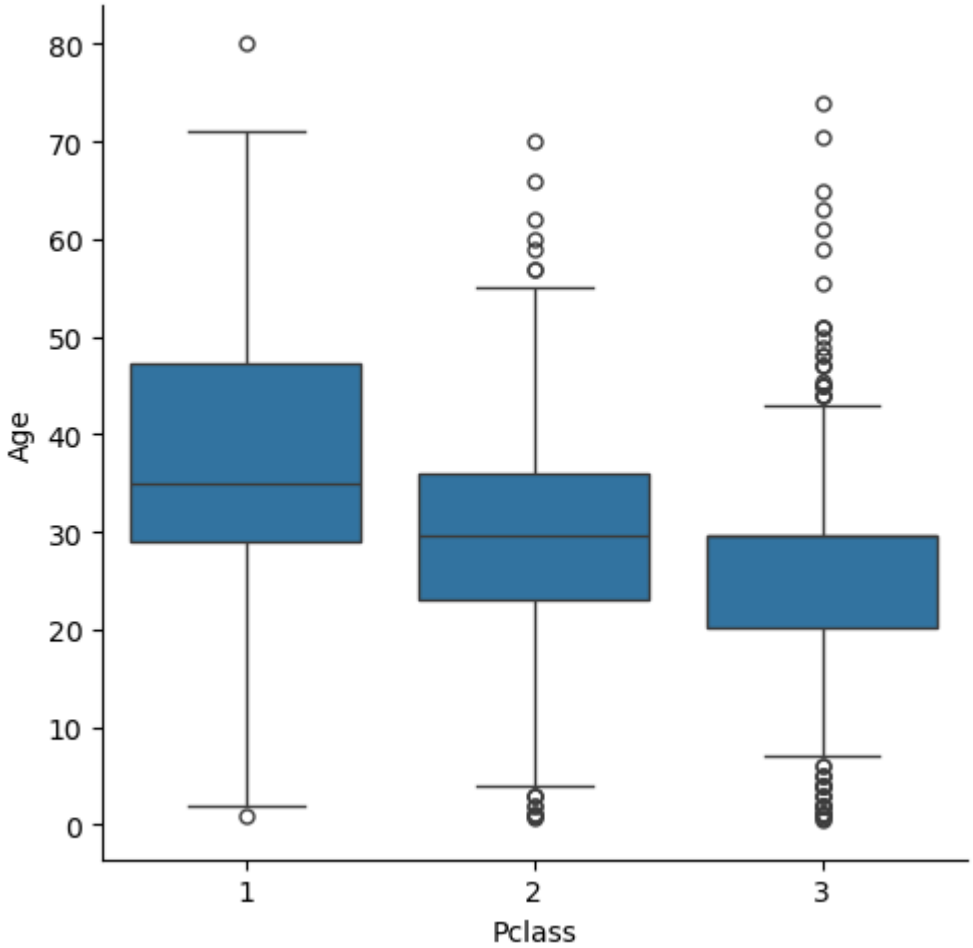
In [30]:
```
sns.boxplot(x='SibSp', data=data)
```

Out[30]:
```
<Axes: xlabel='SibSp'>
```



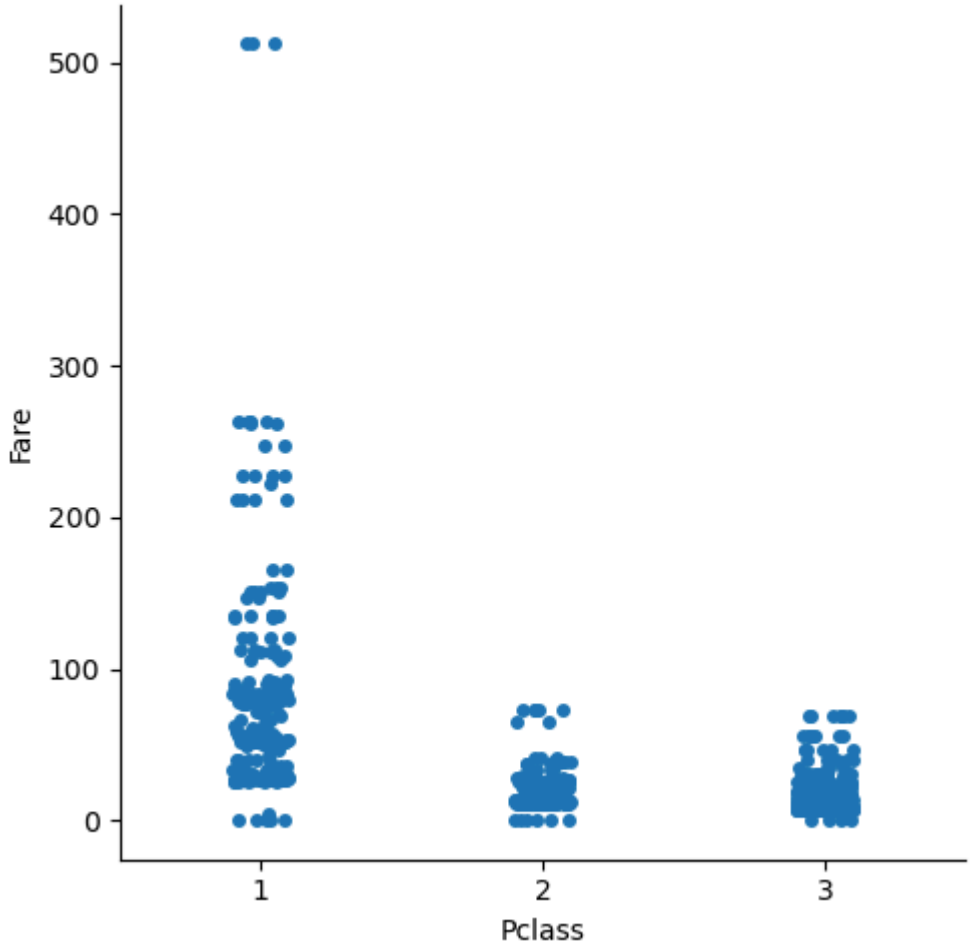
In [31]:
```
sns.catplot(x= 'Pclass', y = 'Age', data=data, kind = 'box')
```

Out[31]: <seaborn.axisgrid.FacetGrid at 0x2788213ba10>



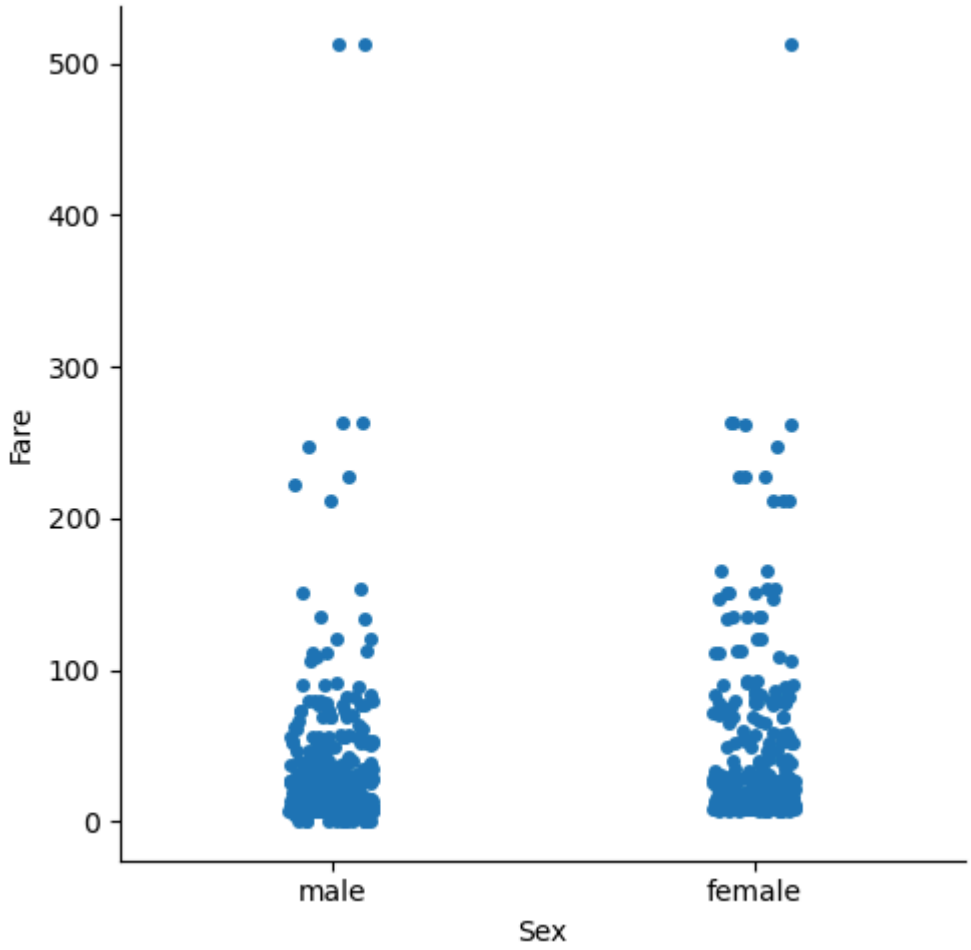
In [32]:
```
sns.catplot(x= 'Pclass', y = 'Fare', data=data, kind = 'strip')
```

Out[32]:
```
<seaborn.axisgrid.FacetGrid at 0x2788d364690>
```



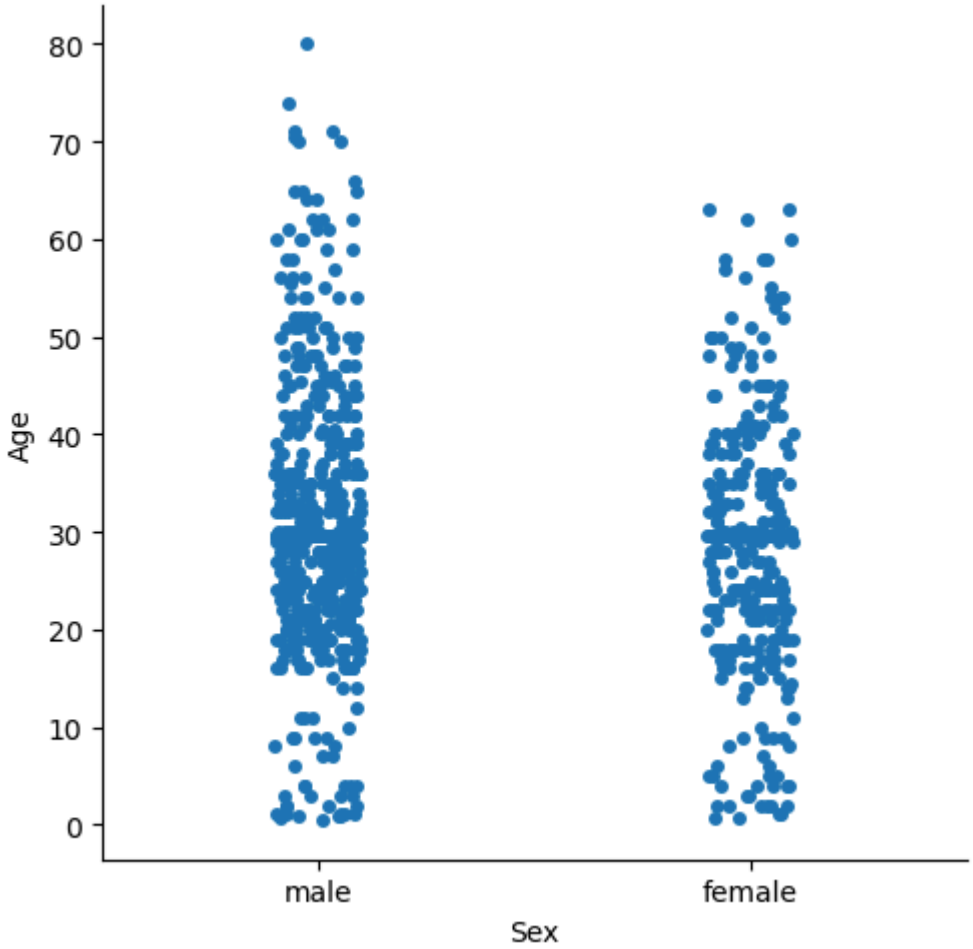
In [33]:
```
sns.catplot(x= 'Sex', y = 'Fare', data=data, kind = 'strip')
```

Out[33]:
```
<seaborn.axisgrid.FacetGrid at 0x2788d3ca350>
```



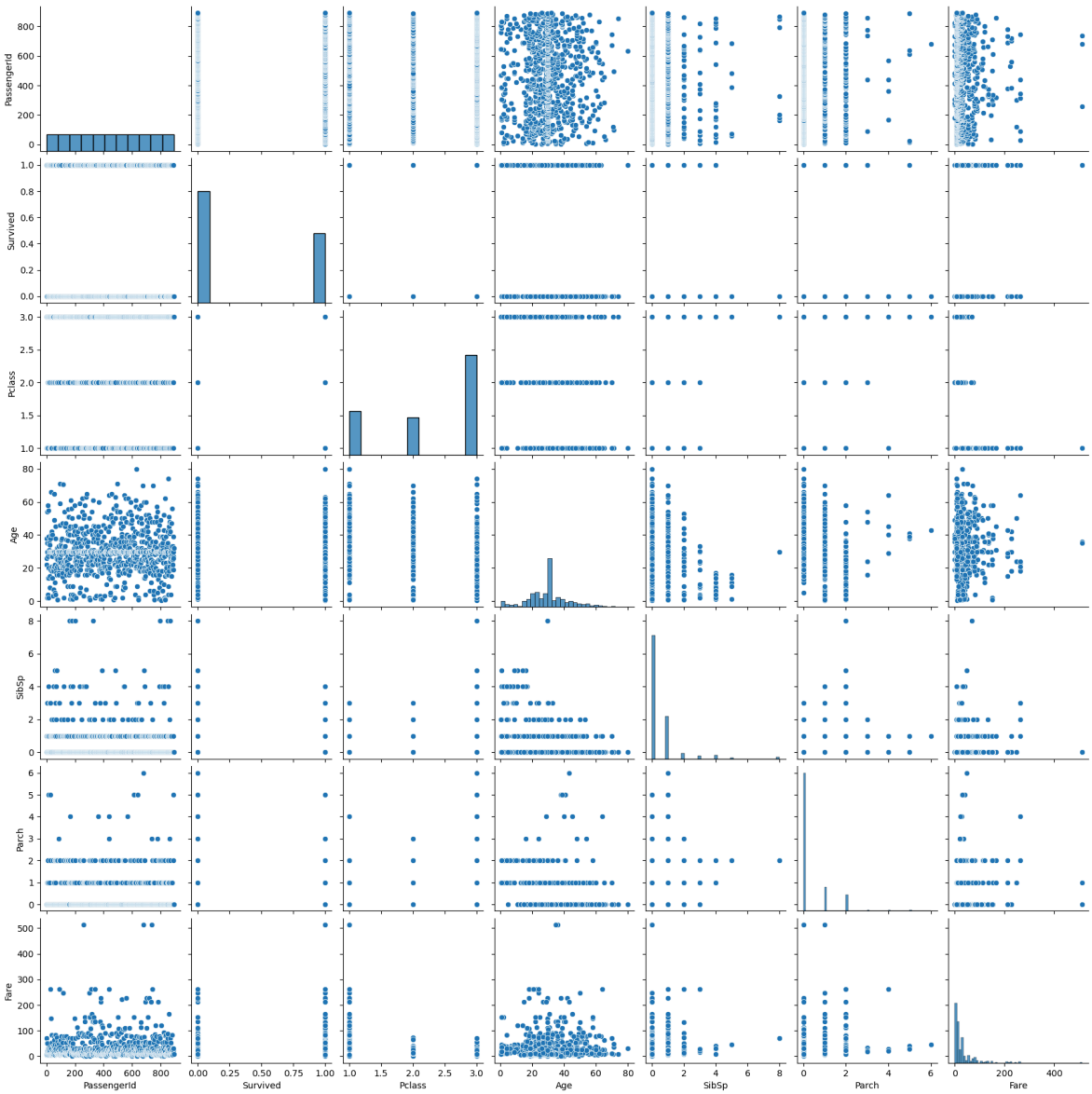
In [34]:
```
sns.catplot(x= 'Sex', y = 'Age', data=data, kind = 'strip')
```

Out[34]:
```
<seaborn.axisgrid.FacetGrid at 0x2788d42f390>
```



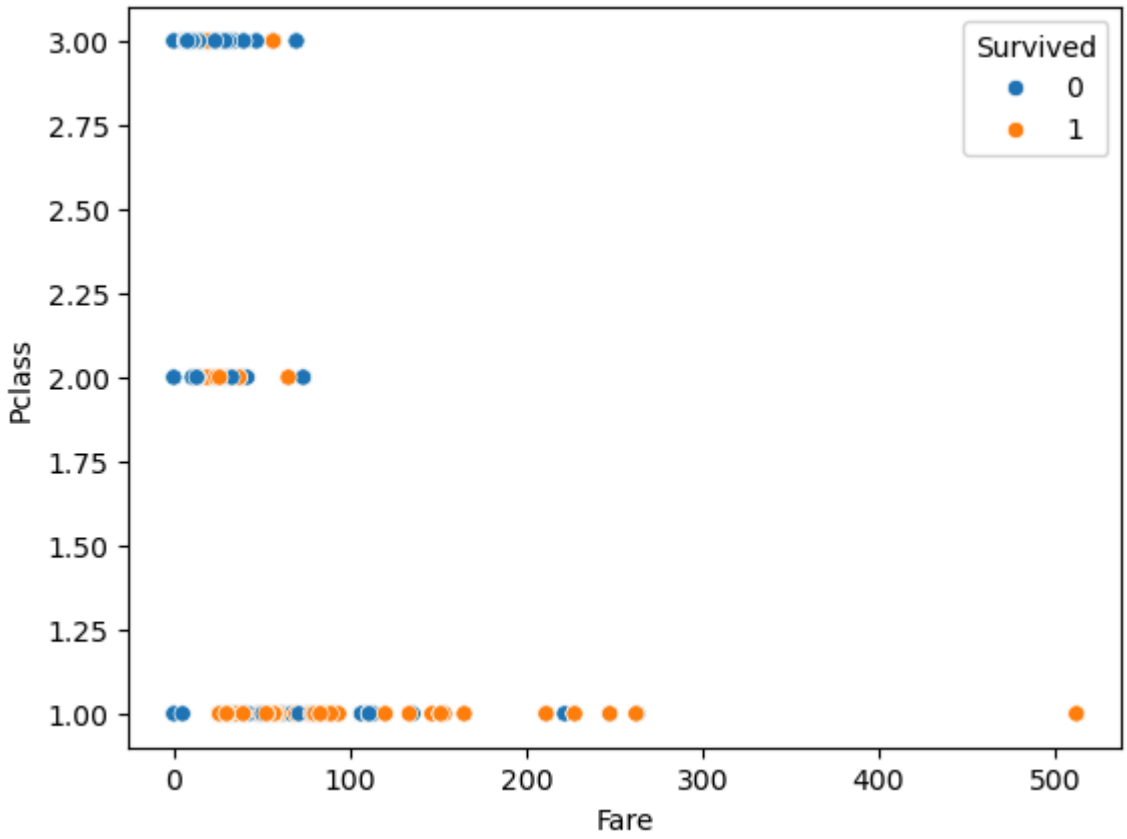
In [36]:
```
sns.pairplot(data)
```

Out[36]:
```
<seaborn.axisgrid.PairGrid at 0x2789061fb10>
```



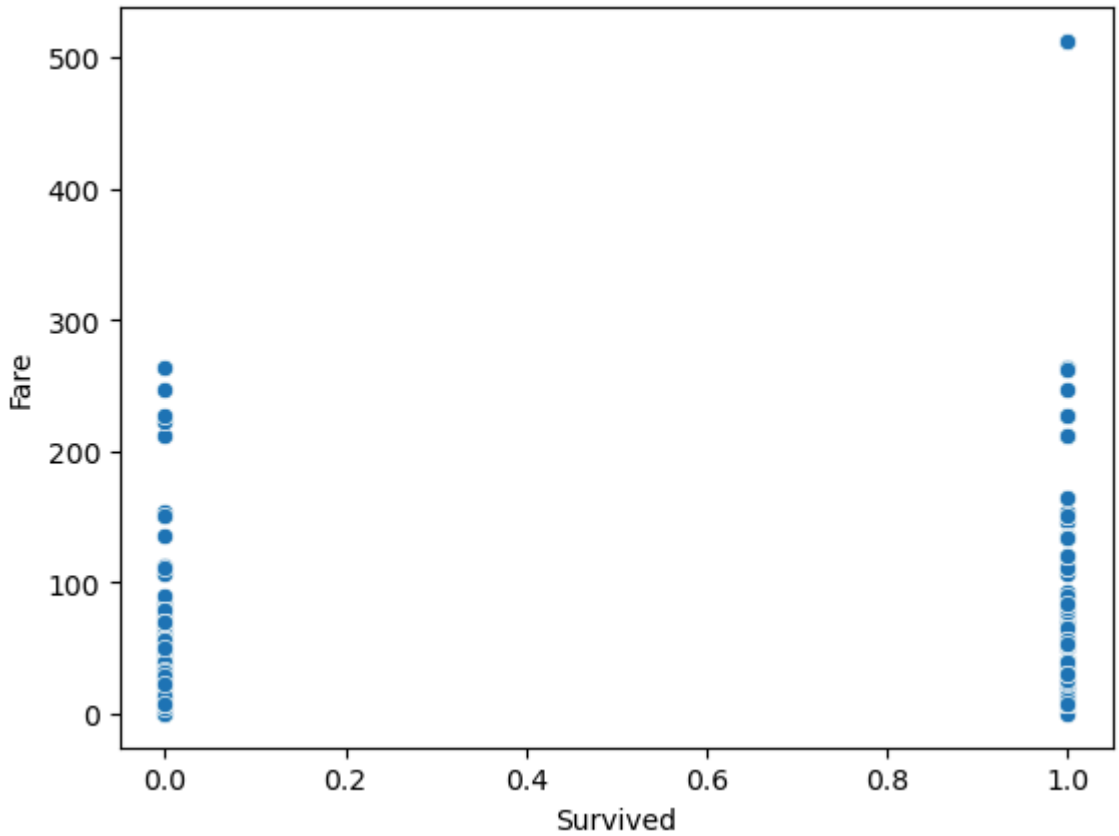
In [37]:
```
sns.scatterplot(x = 'Fare', y = 'Pclass', hue = 'Survived', data = data)
```

Out[37]:
```
<Axes: xlabel='Fare', ylabel='Pclass'>
```



In [38]:
```
sns.scatterplot(x = 'Survived', y = 'Fare', data = data)
```

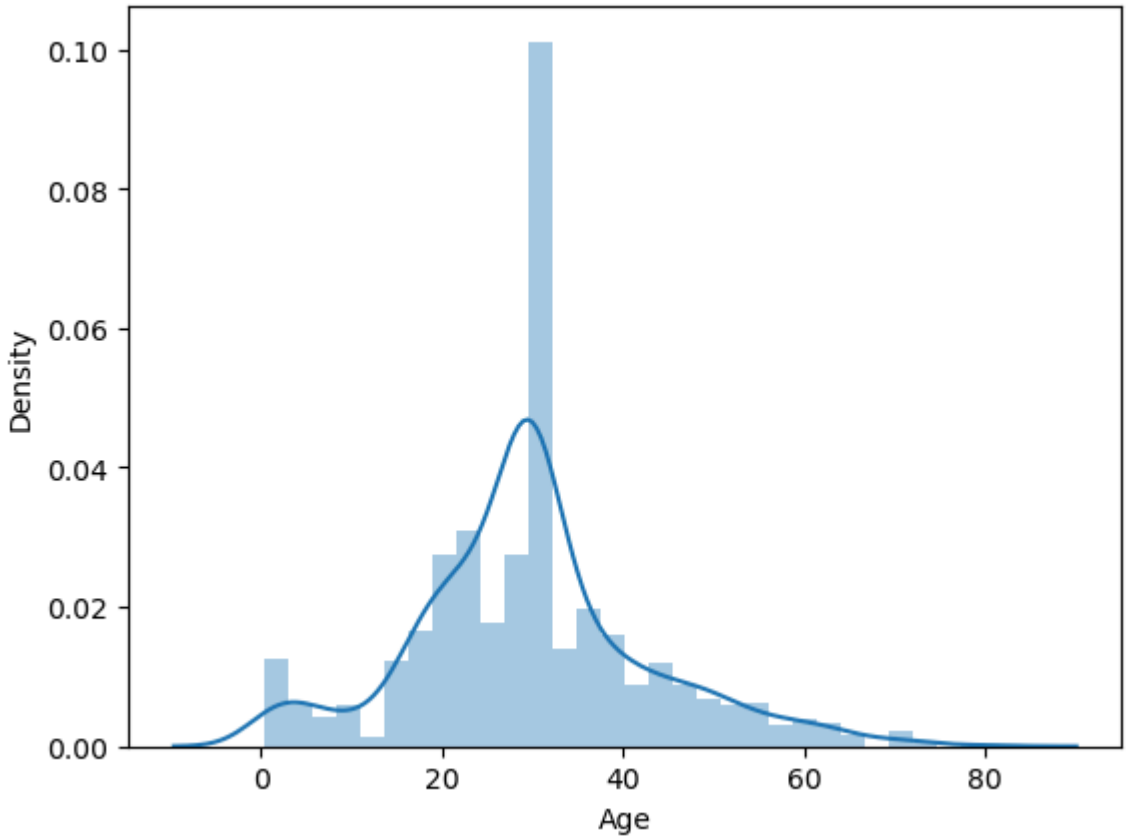
Out[38]: <Axes: xlabel='Survived', ylabel='Fare'>



In [39]:
```
sns.distplot(data['Age'])
```

```
C:\Users\adity\AppData\Local\Temp\ipykernel_5804\2317092479.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(data['Age'])
```

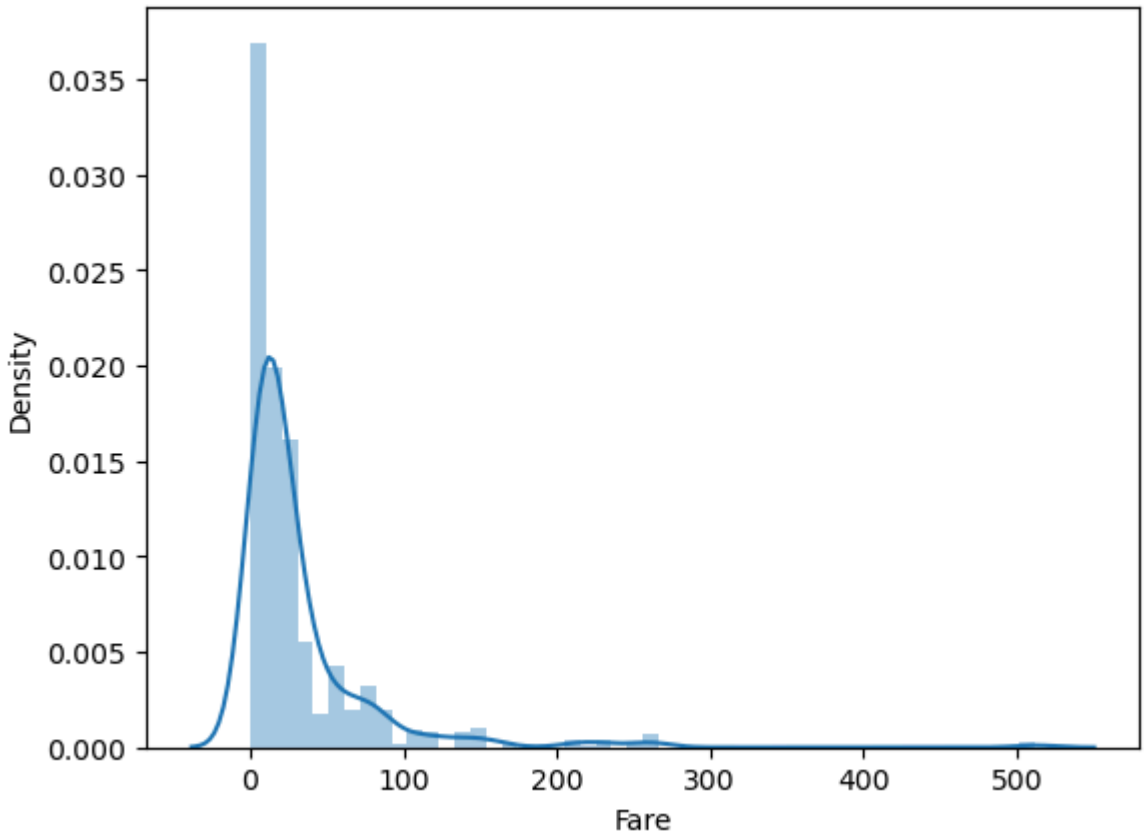
Out[39]: <Axes: xlabel='Age', ylabel='Density'>



In [42]:
```
sns.distplot(data['Fare'])
```

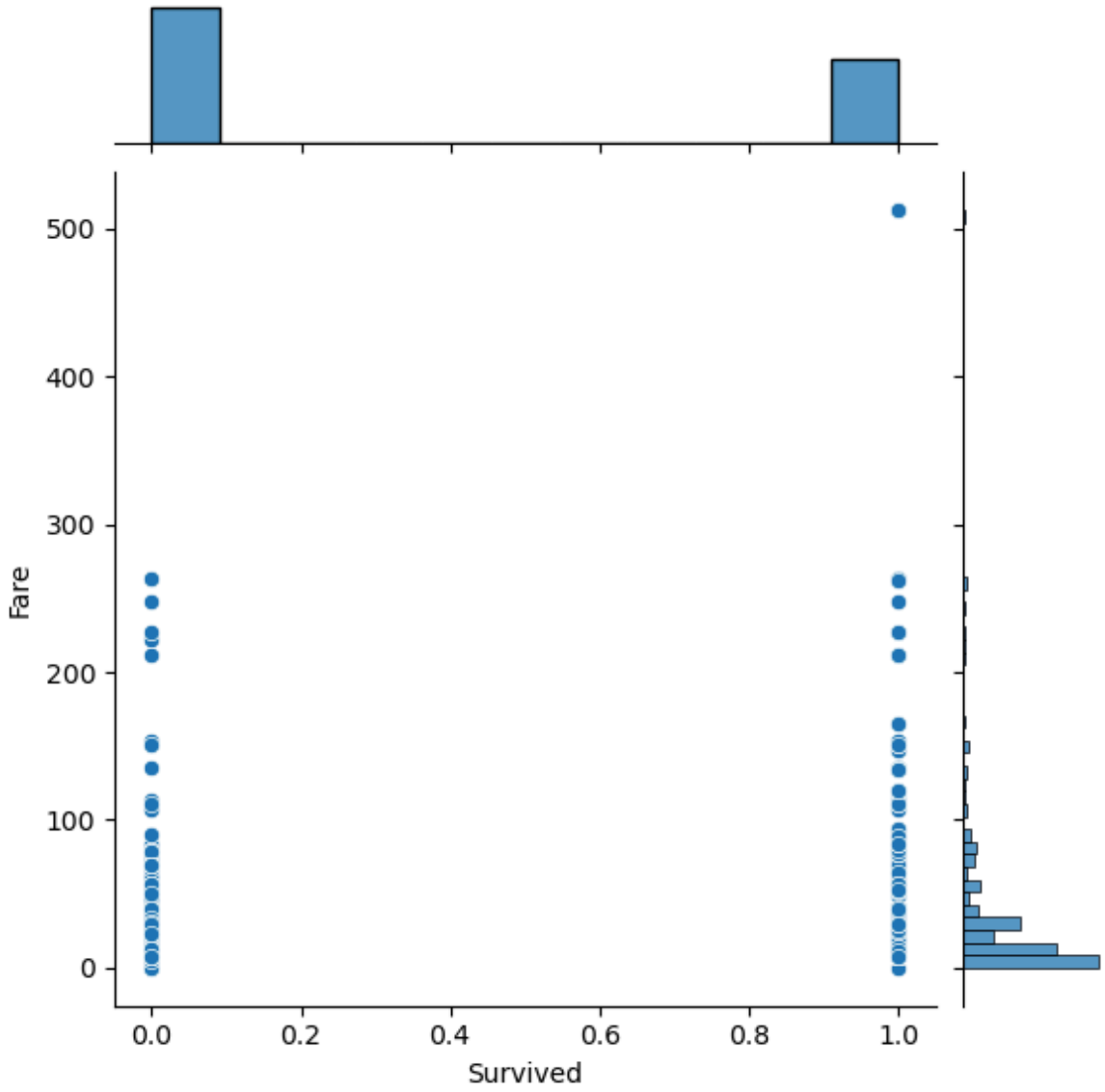
```
C:\Users\adity\AppData\Local\Temp\ipykernel_5804\2921470011.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(data['Fare'])
```

Out[42]: <Axes: xlabel='Fare', ylabel='Density'>



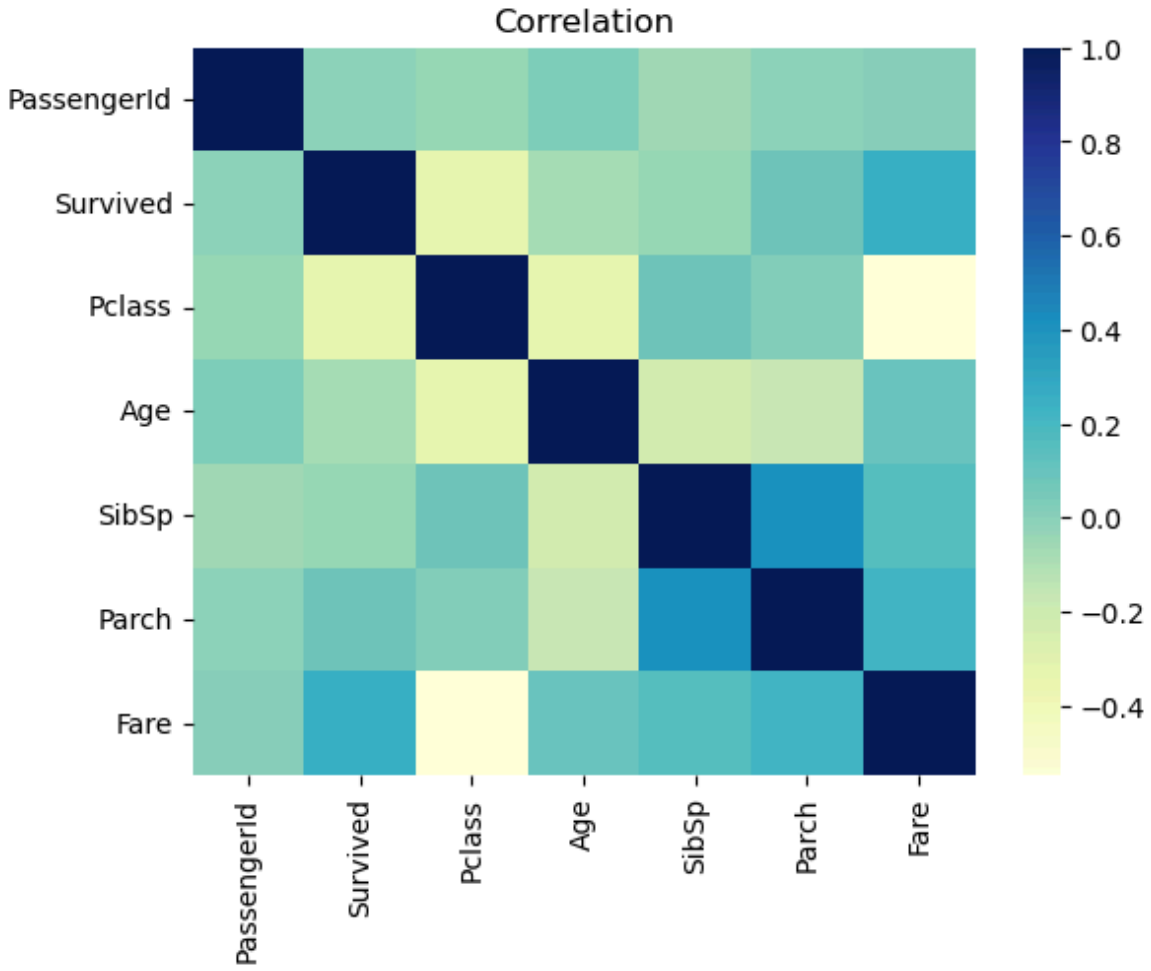
In [43]:
```
sns.jointplot(x = "Survived", y = "Fare", kind = "scatter", data = data)
```

Out[43]:
```
<seaborn.axisgrid.JointGrid at 0x278849d12b0>
```



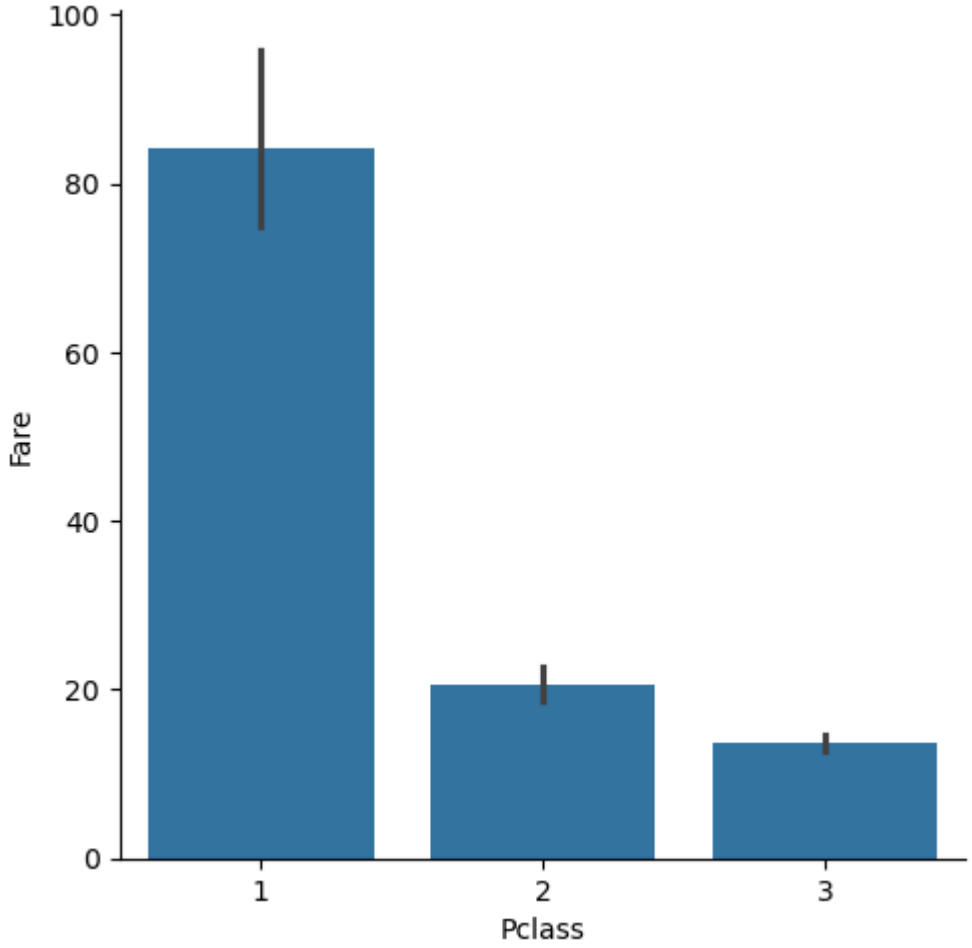
In [47]:
```
tc = data.corr(numeric_only=True)
sns.heatmap(tc, cmap="YlGnBu")
plt.title('Correlation')
```

Out[47]:
```
Text(0.5, 1.0, 'Correlation')
```



In [48]:
```
sns.catplot(x='Pclass', y='Fare', data=data, kind='bar')
```

Out[48]:
```
<seaborn.axisgrid.FacetGrid at 0x278990a3750>
```



In [ ]: